

PYTHON MINI PROJECT

Rule-Based Conversational Chatbot

STUDENT	Kovvuri Charan Sai
REGISTER NUMBER	99230040353
DEPARTMENT	CSE
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LANGUAGE	Python

1. Objective

The objective is to create a simple rule-based chatbot in Python. It should continuously accept messages, identify common user intents using predefined rules, provide an appropriate response, and stop when the user enters an exit command.

2. Approach

The chatbot uses a loop-based conversation model. Each input is converted to lowercase and checked against groups of keywords. Different rules handle greetings, questions about the bot, help requests, and unknown messages. The commands **bye** and **exit** terminate the conversation.

3. Python Implementation

```
def chatbot():
    print("Bot: Hello! I am PyBuddy. Type 'bye' to stop.")

    while True:
        message = input("You: ").strip().lower()

        if message in ("bye", "exit"):
            print("Bot: Goodbye! Have a great day.")
            break

        elif any(word in message for word in ["hello", "hi", "hey"]):
            print("Bot: Hi! Nice to meet you.")

        elif "name" in message:
            print("Bot: I am PyBuddy, a rule-based Python chatbot.")

        elif "help" in message:
            print("Bot: You can greet me, ask my name, or say bye.")

        elif "how are you" in message:
            print("Bot: I am doing well. Thanks for asking!")

        else:
            print("Bot: Sorry, I do not understand that yet.")

chatbot()
```

4. Working of the Chatbot

Step 1 — The program displays an introductory message.

Step 2 — It waits for the user to enter a message.

Step 3 — The message is cleaned and converted to lowercase.

Step 4 — The program checks the message against predefined rules.

Step 5 — A matching response is displayed; otherwise, an unknown-message response is shown.

Step 6 — The loop continues until the user enters **bye** or **exit**.

5. Sample Test Run

```
Bot: Hello! I am PyBuddy. Type 'bye' to stop.  
You: hello  
Bot: Hi! Nice to meet you.  
You: what is your name?  
Bot: I am PyBuddy, a rule-based Python chatbot.  
You: help  
Bot: You can greet me, ask my name, or say bye.  
You: how are you  
Bot: I am doing well. Thanks for asking!  
You: good morning  
Bot: Sorry, I do not understand that yet.  
You: bye  
Bot: Goodbye! Have a great day.
```

6. Result

The rule-based chatbot was successfully designed in Python. It can maintain a simple text conversation, recognize predefined categories of input, respond to supported messages, handle unknown input, and terminate when an exit command is received.

7. Conclusion

This mini project demonstrates the basic idea behind conversational programs using conditional statements, loops, string processing, and keyword matching. Although it does not use machine learning, the structure provides a simple foundation for understanding how chatbot interactions can be implemented.